Homework 3
Statistics 2103

1. A random sample of 9 triathletes had their maximum heart rate measured during a swimming event. The sample mean and standard deviation were 188 and 7.2 beats per minute, respectively. Assume that heart rates are normally distributed.

   (a) Construct a 90% confidence interval for the true mean heart rate of triathletes while swimming.

   (b) To construct the confidence interval in part (a), did you use the Normal distribution, or the t distribution? Justify your choice.

2. Let X be a normally distributed random variable, with mean μ and variance σ^2. What is the distribution of Z, where Z is defined as: $Z = \frac{X-\mu}{\sigma}$? Your response should specify the type of distribution and the value(s) of its parameter(s).

3. A manufacturer of glass bottles redesigns their packaging to try to reduce the proportion of bottles that break during shipping. With the old packaging the probability of a bottle breaking was 8%, and the new probability of breaking is p. The manufacturer is interested in answering whether the new probability of breaking is less than 8%. In a random sample of $n = 300$ bottles shipped in the new packaging, 16 break.

   (a) State an appropriate null and alternative hypothesis for this problem.

   (b) Calculate the p value for this hypothesis test.

   (c) Using a rejection threshold of 0.05, state your conclusion and explain what that conclusion means in the specific context of this problem.

4. A town's drinking water had a uranium concentration of 20 micrograms per liter in 2016. They are interested in testing whether the average uranium concentration in 2026, μ, has changed since 2016. Suppose that current uranium measurements have an unknown mean μ, and known standard deviation $\sigma = 3$. In 40 random water samples, the average uranium concentration is $\bar{x} = 18.9$.

   (a) State an appropriate null and alternative hypothesis for this problem.

   (b) Calculate the p value for this hypothesis test.

   (c) With a rejection threshold of 0.05, what is your conclusion?

(d) What would your conclusion be if using a rejection threshold of 0.01?

5. Suppose we are interested in the relationship between a car's weight and the time it takes to accelerate from 0 to 60 miles per hour. The file `cars.csv` contains data for 15 cars, consisting of the weight (in lbs) and acceleration time (in seconds).

   (a) Make a scatter plot of the car data, with weight on the x axis.

   (b) Find the slope and intercept of the least squares regression line for this data (again, with "weight" as the predictor variable x.)

   (c) What is the expected acceleration time for a car weighing 4315 lbs?